THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 7 & 8: **Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Particles of matter are held together by: 1

(a) friction only **(b)** gravity of Moon
(c) interparticle forces of attraction **(d)** magnetic poles only

2. Solids have a fixed: 1

(a) shape only **(b)** shape and volume **(c)** volume only **(d)** neither

3. The particles in a liquid: 1

(a) are completely fixed **(b)** do not exist
(c) are completely free like gas **(d)** move past each other within a space

4. The temperature at which a solid melts at atmospheric pressure is its: 1

(a) boiling point **(b)** sublimation limit
(c) melting point **(d)** freezing point of vapour

5. The melting point of urea is: 1

(a) 0 °C **(b)** 1538 °C **(c)** 133 °C **(d)** 100 °C

6. Air is a: 1

(a) mineral **(b)** mixture of gases **(c)** compound **(d)** element

7. Alloys are mixtures of: 1

(a) minerals only **(b)** gases only
(c) a metal with other metals or non-metals **(d)** liquids only

8. Kamsya (bronze) had copper and tin in the ratio: 1

(a) 1 : 1 **(b)** 4 : 1 **(c)** 1 : 4 **(d)** 2 : 1

9. An element is a substance that: 1

(a) has many types of atoms
(b) is always a gas
(c) can be split by physical means
(d) cannot be broken down into simpler substances

10. Water has different properties than hydrogen and oxygen because it is a: 1

(a) compound **(b)** mixture **(c)** element **(d)** alloy

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does the smell of perfume reach us quickly? 2

12. What happens to the particles of a solid on heating? 2

13. Write the word equation for heating iron and sulfur. 2

14. What are metalloids? Give two examples. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Why can a gas be filled in a container of any size? 3

16. Write the word equation when magnesium burns in air and classify each substance. 3

17. Why does a candle's wax change state near the flame? Describe. 3

OR

The melting points are ice 0 °C, urea 133 °C and iron 1538 °C. Arrange in order of strength of interparticle forces. 3

18. How can gold be both a metal and a mineral? 3

OR

Name two compounds made of only non-metals and give two uses of each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain why ice floats on water using particles and why this is an exception. 5

OR

Two students argue whether rice flour is a solid or a liquid. Settle the argument with reasons. 5

20. Explain how a mixture of iron filings, sulfur and sand can be separated. 5

OR

A student claims sugar is a mixture because it can be burnt to form carbon and water. Evaluate the claim. 5

END OF QUESTION PAPER — Total: 40 Marks

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